

Question	Marking Guidance	Mark	Comments
10.1	1. Correct answer of 19.4/19.41% OR 19.47/19.5% = 2 marks ;; 2. Incorrect answer but shows increase of 1,048,320 OR 1,051,200 = one mark;	2	Accept: 19.46% for one mark.
10.2	1. Less/no acetylcholine broken down; 2. Acetylcholine attaches to <u>receptors</u> ; 3. (More) Na ⁺ enter to reach threshold/for depolarisation/action potential/impulse;	3	1. Accept: more acetylcholine present/remains. 1 and 2. Accept: remains attached for longer = 2 marks. 3. Must be sodium ions.
10.3	1. Isolated so inbreeding/low genetic diversity/small gene pool; 2. <u>Allele</u> inherited (through generations) from (common) ancestor;	2	1. Ignore: Founder effect. 1. Accept: no interbreeding with other populations. 1. Reject: interbreeding within the population.
10.4	1. AD/symptoms develops late/at 49; 2. Have already reproduced;	2	Note: 'It' is not equivalent to AD/symptom as the question stem relates to the mutation.
10.5	1. Epigenetics/environment/named factor e.g. stress, alcohol, toxins, diet, exercise, smoking; 2. methylation (of genes) OR acetylation (of histones);	2	1. Ignore: gender and lifestyle. 2. If further details are provided the context must be correct e.g. increased methylation or decreased acetylation inhibit gene expression/transcription.
10.6	1. One person was homozygous dominant/has two dominant alleles = 2 marks ;; 2. For one mark has two alleles/chromosomes;	2	1. Accept; homozygous dominant genotype e.g. 'one person has AA' for 2 marks. 2. Accept: is diploid or

			has two copies of the gene.
10.7	1. (GCA/triplet) is common/found in other places; 2. Would not know if it was the mutation/allele/gene OR Produces 'false positives';	2	1. Accept: Probe will bind elsewhere.